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Prove that the force field

$$\vec{F} = (y^2 - 2xyz^3)\hat{i} + (3 + 2xy - x^2z^3)\hat{j} + (6z^3 - 3x^2yz^2)\hat{k}$$

is conservative.

Solution:

We know that he force field is conservative iff curl $\vec{F}$ is zero;

i.e. $\nabla \times \vec{F} = 0$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 - 2xyz^3 & 3 + 2xy - x^2z^3 & 6z^3 - 3x^2yz^2 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(6z^3 - 3x^2yz^2) - \frac{\partial}{\partial z}(3 + 2xy - x^2z^3)\right] - \hat{j}\left[\frac{\partial}{\partial x}(6z^3 - 3x^2yz^2) - \frac{\partial}{\partial z}(y^2 - 2xyz^3)\right]$$

$$+ \hat{k}\left[\frac{\partial}{\partial x}(3 + 2xy - x^2z^3) - \frac{\partial}{\partial y}(y^2 - 2xyz^3)\right]$$

$$= (-3x^2z^2 + 3x^2z^2)\hat{i} - (-6xyz^2 + -6xyz^2)\hat{j} + (2y - 2xz^2 - 2y + 2xz^2)\hat{k}$$

$$= 0$$

As we obtain $\nabla \times \vec{F} = 0$

We conclude that $\vec{F}$ is conservative.

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An example of non-conservative force, we have $F = kv$, where v is the velocity of then,

$$\oint \vec{F}.d\vec{r} = \int \vec{F}.\frac{d\vec{r}}{dt}dt$$

$$= \int_{t_1}^{t_2} \vec{F}.\vec{v}dt = k\int_{t_1}^{t_2} v^2 dt > 0$$

which shows the integral is not equals to zero. Hence the force is non-conservative.

Work Energy relation and Non-conservative Forces

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Example of Non- Conservative Field

Problem:

Show that the force field given by

$$\vec{F} = x^2yz\hat{\imath} - xyz^2\hat{\jmath} + 2xz\hat{k}$$

is non-conservative.

Solution:

We have

$$\vec{F} = x^2yz\hat{\imath} - xyz^2\hat{\jmath} + 2xz\hat{k}$$

To verify whether the force field F is conservative or not, we will check whether $\nabla \times \vec{F} = 0$ or not.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2yz & -xyz^2 & 2xz \end{vmatrix}$$

$$= \hat{\imath}\left[\frac{\partial}{\partial y}(2xz) - \frac{\partial}{\partial z}(-xyz^2)\right] - \hat{\jmath}\left[\frac{\partial}{\partial z}(x^2yz) - \frac{\partial}{\partial x}(2xz)\right] + \hat{k}\left[\frac{\partial}{\partial x}(-xyz^2) - \frac{\partial}{\partial y}(x^2yz)\right]$$

$$= (2xyz)\hat{\imath} - (x^2y - 2z)\hat{\jmath} + (yz^2 + x^2z)\hat{k}$$

$$\neq 0$$

As we obtain $\nabla \times \vec{F} \neq 0$

We conclude that $\vec{F}$ is non-conservative.

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Energy of S.H.M

If T is the kinetic energy, V the potential energy and $E = T + V$ the total energy of a simple harmonic oscillator, then we have

$$K.E = T = \frac{1}{2}mv^2$$

and

$$V = \frac{1}{2}kx^2$$

Then the total energy of S.H.M will be

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

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ENERGY OF A SIMPLE HARMONIC OSCILLATOR

Example:

Find the total energy of the force $\vec{F} = -3xi$ acting on a simple harmonic oscillator, where i represents the direction.

Solution:

Since the total energy of SHM

$$E = T + V$$

By Newton's second law,

$$F = ma$$

therefore

$$F = m\frac{dv}{dt} = -3x$$

$$\frac{dv}{dt} = -\frac{3}{m}x$$

Integrating with respect to t, we have

$$v = -\frac{3}{2m}x^2 + c$$

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$$v = -\frac{3}{2m}x^2 + c$$

c can be calculated if the initial condition is given. Assuming $v = 0$ initially, which gives $c = 0$.

Thus $v = -\frac{3}{2m}x^2$

So,

$$T = \frac{9}{8m}x^4$$

Now, as the potential energy is given by V where $\vec{F} = -\nabla V$

or $F = -3xi = -(\frac{\partial V}{\partial x}i + \frac{\partial V}{\partial y}j + \frac{\partial V}{\partial z}k)$

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Then

$$\frac{\partial V}{\partial x} = 3x\,,\ \frac{\partial V}{\partial y} = 0,\ \frac{\partial V}{\partial z} = 0$$

By integrating, we obtain

$$V = \frac{3}{2}x^2 + c_1$$

Assuming $V = 0$, corresponding to $x = 0$, we get $c_1 = 0$.

So the potential energy is $V = \frac{1}{2}kx^2$

Hence

$$E = \frac{9}{8m}x^4 + \frac{1}{2}kx^2$$

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Problem Statement

A particle of mass 4 moves along y axis attracted towards the origin by a force whose magnitude is numerically equal to $6y$. If the particle is initially at rest at $y = 10$. Consider that the particle has also a damping force whose magnitude is numerically equal to 6 times the instantaneous speed.

Find

i. Position of the particle
ii. Velocity of the particle at any time

Solution

i. Since we have by Newton's second law

$$F = ma = 4\ddot{y}$$

According to the given information, the damping force is $-6\dot{y}$.
So the net force is

$$-6y - 6\dot{y}$$

Hence,

$$4\ddot{y} = -6y - 6\dot{y}$$

or

$$\ddot{y} + \frac{2}{3}\dot{y} + \frac{2}{3}y = 0$$

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or

$$\ddot{y} + \frac{2}{3}\dot{y} + \frac{2}{3}y = 0$$

The solution of the above equation is

$$y = e^{-\sqrt{2/3}t}(A + Bt)$$

and

$$\dot{y} = Be^{-\sqrt{2/3}t} - \sqrt{2/3}\,(A + Bt)e^{-\sqrt{2/3}t}$$

To find the values of constants, we use I. Cs

At $t = 0$, $y = 10$ and $dy/dt = 0$; thus,

$$A = 10$$

and $$0 = B(1) - \sqrt{2/3}\,(10)$$

$$B = 10\sqrt{2/3}$$

and the solution gives

$$y = e^{-\sqrt{2/3}t}(A + Bt)$$
$$= 10e^{-\sqrt{2/3}t}(1 + \sqrt{2/3}\,t)$$

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Module No. 108

Euler's Theorem – Derivation

The following theorem, called Euler's theorem, is fundamental in the motion of rigid bodies.

Theorem Statement

"A rotation of a rigid body about a fixed point of the body is equivalent to a rotation about a line which passes through the point."

The line referred to is called the instantaneous axis of rotation. Rotations can be considered as finite or infinitesimal. Finite rotations cannot be represented by vectors since the commutative law fails. However, infinitesimal rotations can be represented by vectors.

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Module No. 109

Chasles' Theorem

Theorem Statement:

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line (called its screw axis) followed (or preceded) by a rotation about that line.

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Module No. 111

The Concept of Rectilinear Motion of Particles, Uniform Rectilinear Motion, Uniformly Accelerated Rectilinear Motion

When a moving particle remains on a single straight line, the motion is said to be rectilinear. In this case, without loss of generality we can choose the x-axis as the line of motion.
The general equation of motion is then

$$F = ma \Longrightarrow F(x, \dot{x}, \ddot{x}) = m\ddot{x}$$

Rectilinear motion for a particle:

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An example of linear motion is an athlete running along a straight track.

The rectilinear motion can be of two types:

i. Uniform rectilinear motion
ii. Non uniform rectilinear motion

Uniform Rectilinear Motion:

Uniform rectilinear motion is a type of motion in which the body moves with uniform velocity or zero acceleration.

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The time integral of the force F given by

$$\int_{t_1}^{t_2} F dt$$

is called the impulse of the force F.

Theorem Statement

The impulse is equal to the change in momentum; or, in symbols,

$$\int_{t_1}^{t_2} F dt = mv_2 - mv_1 = p_2 - p_1$$

Proof

We have to prove that the impulse of a force is equal to the change in momentum.
By definition of impulse and Newton's second law, we have

$$\int_{t_1}^{t_2} F dt = \int_{t_1}^{t_2} m\frac{dv}{dt} dt = \int_{t_1}^{t_2} m dv = m|v|_{t_1}^{t_2} = mv_2 - mv_1 = p_2 - p_1$$

where we use the conditions
$v(t_1) = v_1$ and $v(t_2) = v_2$
The theorem is true even when the mass is variable and the force is non-conservative.

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Example:
What is the magnitude of the impulse developed by a mass of 200 gm which changes its velocity from $5i - 3j + 7k$ m/sec to $2i + 3j + k$ m/sec?

Solution:
Since we have the following information:

$$m = 200 \text{ gm}$$
$$v_1 = 5i - 3j + 7k$$
$$v_2 = 2i + 3j + k$$

As we know that

$$\text{Impulse} = p_2 - p_1$$
$$= mv_2 - mv_1$$
$$= m(v_2 - v_1)$$

Substituting the values, we get

$$\text{Impulse} = 200(2i + 3j + k - (5i - 3j + 7k))$$
$$= 200(-3i + 6j - 6k)$$

The magnitude of the Impulse will be

$$\text{Impulse magnitude} = 200\sqrt{9 + 36 + 36}$$
$$= 200\sqrt{81}$$
$$= 200(9)$$
$$= 1800\text{mgm/sec}$$
$$= 1.8 \text{ N sec}$$

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Module No. 119

Torque

Definition

Torque is defined as the turning effect of a body. It is trend of an acting force due to which the rotational motion of a body changes. It is also called twist and rotational force on an object. Mathematically, torque is defined as the cross product of the force vector to the distance vector, which causes rotational motion of the body.

$$\boldsymbol{\tau} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{F}}$$

The magnitude of torque depends upon the applied force, the length of the lever arm connecting the axis to the point where the force applied, and the angle between the force vector and the length of lever arm. Symbolically we can write it as:

$$\boldsymbol{\tau} = |\boldsymbol{r}||\boldsymbol{F}| \sin \boldsymbol{\theta}$$

Torque is a vector quantity implies that it has direction as well as magnitude.
The SI unit for torque is the newton meter (Nm).
The direction of torque can be approximate using **Right Hand Rule.**

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Theorem:

The torque acting on a particle equals the time rate of change in its angular momentum, i.e.,

$$\tau = \frac{d\Omega}{dt}$$

where $\Omega = r \times p$ is defined angular momentum of the system.

Proof:

As we know torque is defined as

$$\tau = r \times F = r \times ma$$

as m is constant, therefore

$$= m(r \times a)$$
$$= m(r \times \frac{dv}{dt})$$
$$= m\frac{d}{dt}(r \times v)$$
$$= \frac{d}{dt}(r \times mv)$$
$$= \frac{d}{dt}(r \times p)$$

where $p = mv$ is defined as linear momentum and hence the theorem.

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Product of Inertia

The product of inertia for the same particle w.r.to the pair of coordinate axes are defined as

$$I_{xy} = -mxy$$

$$I_{yz} = -myz$$

$$I_{zx} = -mzx$$

These definitions can be easily generalized to a system of particle and a rigid body.

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Example:

Find the radius of gyration, K, of the triangular lamina of mass M and moment of inertia $I = {}^{1}/_{6}\,Mh^2$.

Solution:

Since formula for radius of gyration is given by

$$K^2 = \frac{I}{M}$$

by substituting value in above , we get

$$K^2 = \frac{{}^{1}/_{6}\,Mh^2}{M} = {}^{1}/_{6}\,h^2$$

or

$$K = \frac{h}{\sqrt{6}}$$

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Given Data

Mass of particle $A = m_1 = 6\text{kg}$

Velocity of A before collision $= u_1 = 5\text{ms}^{-1}$

Velocity of A after collision $= v_1 = 1.5\text{ms}^{-1}$

Mass of Particle $B = m_2 = 4\text{kg}$

Velocity of B before collision $= u_2 = 3\text{ms}^{-1}$

Required

Velocity of B after collision $= v_2 = ?$

Solution

It is always useful to depict the collision with the velocities both before and after.
Using the principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$
$$6 \times 5 + 4 \times (-3) = 6 \times 1.5 + 4 \times v_2$$
$$9 = 4 \times v_2$$
$$v_2 = {}^{9}/_{4} = 2.3\ m\ s^{-1}$$

So after the collision particle B moves with a speed of 2.3 m s−1, in the same direction as A.

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Given Data

Mass of particle A= $m_1 = 8\text{kg}$
Velocity of A before collision= $u_1 = (-1i + 4j)\ m\ s^{-1}$
Velocity of A after collision= $v_1 = (-2i + 2j)\ m\ s^{-1}$
Velocity of B before collision= $u_2 = (-0.8i + 1.4j)\ m\ s^{-1}$
Velocity of B after collision= $v_2 = ?3j\ m\ s^{-1}$

Required

Mass of Particle B= $m_2 = ?$

Solution

It is always useful to depict the collision with the velocities both before and after.
Using the principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$6 \times 5 + 4 \times (-3) = 6 \times 1.5 + 4 \times v_2$$

$$9 = 4 \times v_2$$

$$v_2 = {}^{9}/_{4} = 2.3\text{ms}^{-1}$$

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Given Data

Mass of particle A= $m_1 = 8\text{kg}$
Velocity of A before collision= $u_1 = (-1i + 4j)\ m\ s^{-1}$
Velocity of A after collision= $v_1 = (-2i + 2j)\ m\ s^{-1}$
Velocity of B before collision= $u_2 = (-0.8i + 1.4j)\ m\ s^{-1}$
Velocity of B after collision= $v_2 = ?3j\ m\ s^{-1}$

Required

Mass of Particle B= $m_2 = ?$

Solution

It is always useful to depict the collision with the velocities both before and after.
Using the principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$6 \times 5 + 4 \times (-3) = 6 \times 1.5 + 4 \times v_2$$

$$9 = 4 \times v_2$$

$$v_2 = {}^{9}/_{4} = 2.3\text{ms}^{-1}$$

So after the collision particle B moves with a speed of 2.3 ms^{-1}, in the same direction as A.

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Mass of particle $A = m_1 = 8\text{kg}$
Velocity of A before collision $= u_1 = (-1i + 4j)\text{ms}^{-1}$
Velocity of A after collision $= v_1 = (-2i + 2j)\text{ms}^{-1}$
Velocity of B before collision $= u_2 = (-0.8i + 1.4j)\text{ms}^{-1}$
Velocity of B after collision $= v_2 = 3j\ \text{ms}^{-1}$

Required

Mass of Particle $B = m_2 = ?$

Solution

By making use of principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$8(-1i + 4j) + m2(-0.8i + 1.4j) = 8(-2i + 2j) + m_2(3j)$$

$$-8i + 32j + 16i - 16j + m2(-0.8i + 1.4j) - m_2(3j) = 0$$

$$8i + 16j + m_2(-0.8i - 1.6j) = 0$$

$$8i + 16j - m_2(+0.8i + 1.6j) = 0$$

$$8i + 16j = m_2(0.8i + 1.6j)$$

$$m_2 = \frac{0.8i + 1.6j}{8i + 16j}$$

By rationalizing and solving the above fraction, we obtain

$$m_2 = 10kg$$

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Module No. 143

Introduction to Special Moments of Inertia

In this module, we will study some special moment of inertia. In order to introduce the moment of inertia of some specific shapes and special rigid bodies, we assume that the rigid body is of uniform density of particles.

Solid Circular Cylinder

We assume the radius of cylinder is a and mass M about axis of cylinder.

$$I = \frac{1}{2}Ma^2$$

Hollow Circular Cylinder

We assume the radius of cylinder is a and mass M about axis of cylinder.
We consider the thickness of Wall of cylinder is negligible.

$$I = Ma^2$$

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Solid Sphere

We assume the radius of sphere is a and mass M about a diameter.

$$I = \frac{2}{5}Ma^2$$

Hollow Sphere

We assume the radius of sphere is a and mass M about a diameter.
We consider the thickness of sphere is negligible.

$$I = Ma^2$$

Rectangular Plate

We consider sides of length a and b, and mass M about an axis perpendicular to the plate through the center of mass.

$$I = \frac{1}{12}M(a^2 + b^2)$$

Thin Rod

We assume the length of rod is a and mass M about an axis perpendicular to the rod through the center of mass

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$$I = \frac{1}{12}M(a^2 + b^2)$$

Thin Rod

We assume the length of rod is $\boldsymbol{a}$ and mass M about an axis perpendicular to the rod through the center of mass.

$$I = \frac{1}{12}Ma^2$$

Triangular Lamina

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Triangular Lamina

We assume the height ***h*** and mass **M** of lamina.

$$I = \frac{1}{6}Mh^2$$

Right Circular Cone

We assume the radius of the circular cone is a and mass M.

$$I = \frac{3}{10}Ma^2$$

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Particles of masses $2m$, $3m$ and $4m$ are held in a rigid light framework at points $(0,1,1)$, $(1,1,0)$ and $(-1,\ 0,\ 1)$ resp.

Show that the M. I. of the system about the $x, y, z-$ axis are $11,\ 13,\ 12$ respectively.

Solution

Since the masses are 2, 3 and 4 i.e. $m_1 = 2,\ m_2 = 3$ and $m_3 = 4$, and the points are given by $P_1(0,1,1), P_2(1,1,0)$ and $P_3(-1,0,1)$.

We are required to find out the moment of inertia about x-axis, y-axis and z-axis i.e. I_{xx}, I_{yy}, I_{zz}.

$$I_{xx} = \sum_i m_i(y_i^2 + z_i^2)$$

$$= m_1(y_1^2 + z_1^2) + m_2(y_2^2 + z_2^2) + m_3(y_3^2 + z_3^2)$$

$$= 2(1+1) + 3(1+0) + 4(1+0)$$

$$= 4 + 3 + 4 = 11$$

So

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So,

$$I_{xx} = 11$$

$$I_{yy} = \sum_i m_i(x_i^2 + z_i^2)$$

$$= m_1(x_1^2 + z_1^2) + m_2(x_2^2 + z_2^2) + m_3(x_3^2 + z_3^2)$$

$$= 2(0 + 1) + 3(1 + 0) + 4(1 + 1)$$

$$= 2 + 3 + 8 = 1$$

So,

$$I_{yy} = 13$$

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$$I_{zz} = \sum_i m_i(x_i^2 + y_i^2)$$

$$= m_1(y_1^2 + x_1^2) + m_2(x_2^2 + y_2^2) + m_3(x_3^2 + y_3^2)$$

$$= 2(0 + 1) + 3(1 + 1) + 4(1 + 0)$$

$$= 2 + 6 + 4 = 12$$

So,

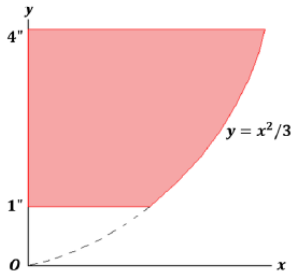
$$I_{zz} = 12$$

- Hence Showed

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Calculate the moment of inertia of the shaded area given in figure about y-axis.



Solution

Here, we have

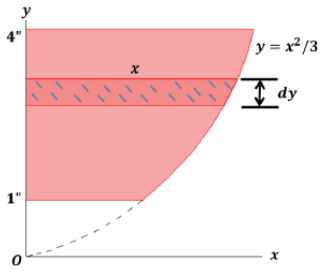
$y = x^2/3$ (1)

We consider an elementary strip from the shaded area whose M.I is

$$I_{strip} = \frac{1}{3}x^2(xdy) = \frac{1}{3}x^3dy$$

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Then by integration, we obtain the moment of inertia of the whole shaded area

$$I_y = \sqrt{3}\int_1^4 y^{3/2}dy = \sqrt{3}\frac{2}{5}\left|y^{5/2}\right|_1^4$$

$$= \sqrt{3}\frac{2}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}\left[(4)^{5/2} - (1)^{5/2}\right]$$

$$= \frac{2\sqrt{3}}{5}(32 - 1)$$

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In order to calculate the moment of inertia through the center of mass c.m., we use parallel axes theorem.

In a transparent notation

$$I_l = I_{cm} + Md^2 \qquad (1)$$

where d is the distance between the origin and the center of mass and $d = {}^{l}/_{2}$.

Also I_l is the moment of inertia of rod about one of its end (which we calculated earlier).

Here

$$I_l = \frac{1}{3}Ml^2 \qquad (2)$$

From (1), we have

$$I_{cm} = I_l - Md^2$$

Substituting value from (2) in (1)

$$I_{cm} = \frac{1}{3}Ml^2 - M\left(\frac{l}{2}\right)^2 = \frac{1}{3}Ml^2 - \frac{1}{4}Ml^2$$

$$I_{cm} = \frac{1}{12}Ml^2$$

which is required moment of inertia about its center of mass.

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Problem Statement

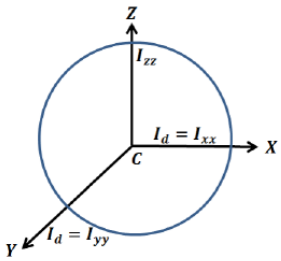
Find the moment of inertia of a uniform circular plate (or disc) about any diameter.

Proof

Since we have deduced that moment of inertia of a uniform circular disc is

$$I_{disc} = \frac{1}{2} Ma^2$$

which is about a line passing through center and perpendicular to the plane.



Considering the same axis with same terms in 3-dim body (i.e. $z - axis$ passing through center

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$$I_{zz} = I_{xx} + I_{yy} \qquad (1)$$

We have to find out M.I. of uniform circular plate (disc) about any of its diameters which are along $x - axis$ and $y - axis$ (i.e. we have to find out I_{xx} *or* I_{yy} both of them are equal to I_d)

Since

$I_{xx} = I_d = I_{yy}$

So equation (1) becomes

$$I_{zz} = I_d + I_d$$

or

$I_{zz} = 2I_d$

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or

$I_{zz} = 2I_d$

$$I_d = \frac{1}{2} I_{zz}$$

$$I_d = \frac{1}{2}\left(\frac{1}{2} \mathrm{M}a^2\right)$$

$(\therefore I_{zz} = I_{disc} = \frac{1}{2} \mathrm{M}a^2)$

$$I_d = \frac{1}{4} M a^2$$

where M is the mass of disc and a is its radius.

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